

=== GNB (1780740597) ===

This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to it.

This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 20 MHz bandwidth.

To run the srsRAN Project gNB with this config, use the following command:

sudo ./gnb -c gnb_zmq.yaml

cu_cp:

amf:

addr: 10.53.1.2 # The address or hostname of the AMF.
port: 38412
bind_addr: 10.53.1.1 # A local IP that the gNB binds to for traffic from the AMF.
supported_tracking_areas:
- tac: 7
plmn_list:
- plmn: "21401"
tai_slice_support_list:
- sst: 1

inactivity_timer: 7200 # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds. Supported: [1 - 7200].

ru_sdr:

device_driver: zmq # The RF driver name.
device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6 # Optionally pass arguments to the selected RF driver.
srate: 11.52 # RF sample rate might need to be adjusted according to selected bandwidth.
tx_gain: 75 # Transmit gain of the RF might need to adjusted to the given situation.
rx_gain: 75 # Receive gain of the RF might need to adjusted to the given situation.

cell_cfg:

dl_arfcn: 632000 # ARFCN of the downlink carrier (center frequency).
band: 78 # The NR band.
channel_bandwidth_MHz: 20 # Bandwidth in MHz. Number of PRBs will be automatically derived.
common_scs: 15 # Subcarrier spacing in kHz used for data.
plmn: "21401" # PLMN broadcasted by the gNB.
tac: 7 # Tracking area code (needs to match the core configuration).
pdcch:
common:
ss0_index: 0 # Set search space zero index to match srsUE capabilities
coreset0_index: 6 # Set search CORESET Zero index to match srsUE capabilities
dedicated:
ss2_type: common # Search Space type, has to be set to common
dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
prach:
prach_config_index: 1 # Sets PRACH config to match what is expected by srsUE
pdsch:
mcs_table: qam64 # Sets PDSCH MCS to 64 QAM
pusch:
mcs_table: qam64 # Sets PUSCH MCS to 64 QAM

log:

filename: /tmp/gnb.log # Path of the log file.
all_level: info # Logging level applied to all layers.
hex_max_size: 0

pcap:

mac_enable: false # Set to true to enable MAC-layer PCAPs.
mac_filename: /tmp/gnb_mac.pcap # Path where the MAC PCAP is stored.
ngap_enable: false # Set to true to enable NGAP PCAPs.
ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
e2ap_enable: true # Set to true to enable E2AP PCAPs.
e2ap_du_filename: /tmp/gnb_du_e2ap.pcap # Path where the DU E2AP PCAP is stored.
e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

```
#e2:
#  enable_du_e2: true           # Enable DU E2 agent (one for each DU instance)
#  e2sm_kpm_enabled: true       # Enable KPM service module
#  e2sm_rc_enabled: true        # Enable RC service module
#  addr: 127.0.0.1              # RIC IP address
#  bind_addr: 127.0.0.100       # A local IP that the E2 agent binds to for traffic from the RIC. ONLY
                                # required if running the RIC on a separate machine.
#  port: 36421                  # RIC port

metrics:
  layers:
    enable_rlc: true            # Enable RLC metrics
    enable_sched: true          # Enable DU scheduler metrics
  periodicity:
    du_report_period: 1000      # Set DU statistics report period to 1s
```

=== UE (1780740597) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 78
nof_carriers = 1

max_nof_prb = 106
nof_prb = 106

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 00112233445566778899aabbccddeeff
imsi = 214010000000001
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER (1780740597) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
# srsRAN is free software: you can redistribute it and/or modify
# it under the terms of the GNU Affero General Public License as
# published by the Free Software Foundation, either version 3 of
# the License, or (at your option) any later version.
#
# srsRAN is distributed in the hope that it will be useful,
# but WITHOUT ANY WARRANTY; without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
# GNU Affero General Public License for more details.
#
# A copy of the GNU Affero General Public License can be found in
# the LICENSE file in the top-level directory of this distribution
# and at http://www.gnu.org/licenses/.
#
```

services:

```
5gc:
  container_name: open5gs_5gc
  build:
    context: open5gs
    target: open5gs
    args:
      OS_VERSION: "22.04"
      OPEN5GS_VERSION: "v2.7.0"
  environment:
    - MONGODB_IP=127.0.0.1
    - OPEN5GS_IP=10.53.1.2
    - UE_IP_BASE=10.45.0
    - UPF_ADVERTISE_IP=10.53.1.2
    - DEBUG=false
```

SUBSCRIBER_DB=214010000000001,00112233445566778899aabbccddeeff,opc,63BFA50EE6523365FF14C1F45F88737D,8000,9,10.45.1.2

```
-
  - NETWORK_NAME_FULL=srsRAN
  - NETWORK_NAME_SHORT=srsRAN
  - TZ=Europe/Madrid
privileged: true
ports:
  - "9898:9999/tcp"
# Uncomment port to use the 5gc from outside the docker network
# - "38412:38412/sctp"
# - "2152:2152/udp"
command: 5gc -c open5gs-5gc.yml
healthcheck:
  test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
  interval: 3s
  timeout: 1s
  retries: 60
networks:
  ran:
    ipv4_address: ${OPEN5GS_IP:-10.53.1.2}
```

```
gnb:
  container_name: srsran_gnb
# Build info
image: srsran/gnb
build:
  context: ..
  dockerfile: docker/Dockerfile
```

```

    args:
      OS_VERSION: "24.04"
# privileged mode is required only for accessing usb devices
privileged: true
# Extra capabilities always required
cap_add:
  - SYS_NICE
  - CAP_SYS_PTRACE
volumes:
  # Access USB to use some SDRs
  - /dev/bus/usb/:/dev/bus/usb/
  # Sharing images between the host and the pod.
  # It's also possible to download the images inside the pod
  - /usr/share/uhd/images:/usr/share/uhd/images
  # Save logs and more into gnb-storage
  - gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current network configuration creates a private network with both containers attached
# An alternative would be `network: host`. That would expose your host network into the container. It's
the easiest to use if the 5gc is not in your PC
networks:
  ran:
    ipv4_address: ${GNB_IP:-10.53.1.3}
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
gnb_config.yml:
  file: ${GNB_CONFIG_PATH:-../configs/gnb_zmq_band78_20mhz.yml} # Path to your desired config file
gnb_compose_config.yml:
  content: |
    cu_cp:
      amf:
        addr: ${OPEN5GS_IP:-10.53.1.2}
        bind_addr: 0.0.0.0
    metrics:
      autostart_stdout_metrics: true
      enable_json: true
    remote_control:
      bind_addr: 0.0.0.0
      enabled: true

volumes:
  gnb-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24
  metrics:
    ipam:
      driver: default
      config:
        - subnet: 172.19.1.0/24

```

=== NOTAS (1780740597) ===

- PLMN (21401) consistente en gNB, UE y SUBSCRIBER_DB (Regla 1).
- dl_arfcn (632000) dentro del rango para Banda 78 (Regla 2).
- common_scs (15) es el mismo que el ssb_scs por defecto (Regla 3).
- Puertos ZMQ cruzados entre gNB y UE (Regla 4).
- bind_addr en gnb_compose_config es 0.0.0.0 (Regla 5).
- Bloque ru_sdr copiado exactamente del CAG (Regla 6).
- IMSI, K y OPC consistentes y en el orden correcto en SUBSCRIBER_DB y UE (Regla 7).
- OPC tomado del template ue_zmq.conf, ya que el usuario no lo especificó.
- El archivo gnb_config.yml se asume que estará en '../configs/gnb_zmq_band78_20mhz.yml'